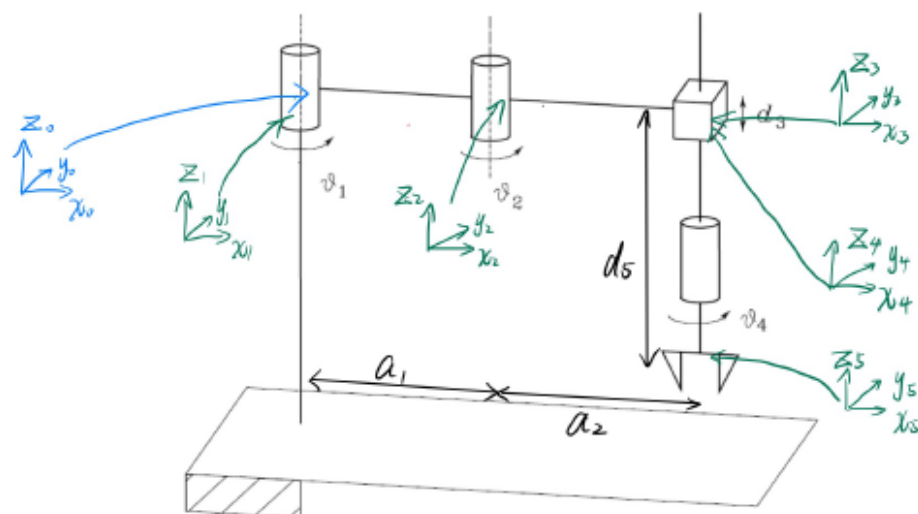


第一次作业答案

1.



如图所示, 设出相应的参数, a_1, a_2, d_3 , 建立相应的坐标系, 得到参数表

	a_{i-1}	a_i	d_i	θ_i
1	0	0	0	θ_1
2	0	a_1	0	θ_2
3	0	a_2	d_3	0
4	0	0	0	θ_4
5	0	0	d_5	0

从而算出变换矩阵

$${}^0T_1 = \begin{pmatrix} C\theta_1 & -S\theta_1 & 0 & 0 \\ S\theta_1 & C\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} C\theta_2 & -S\theta_2 & 0 & a_1 \\ S\theta_2 & C\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

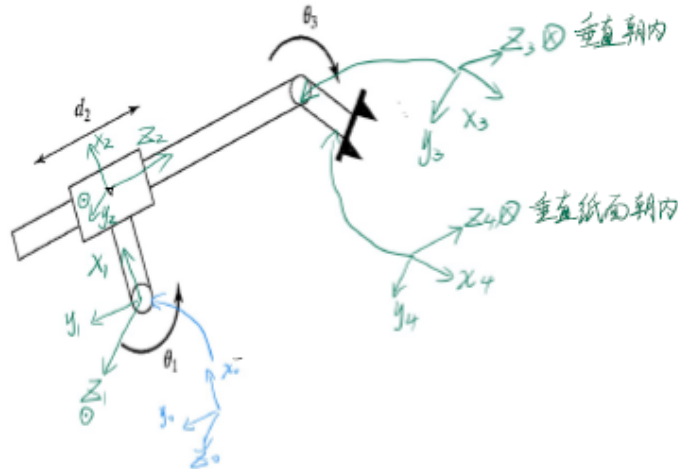
$${}^2T_3 = \begin{pmatrix} 1 & 0 & 0 & a_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^3T_4 = \begin{pmatrix} C\theta_4 & -S\theta_4 & 0 & 0 \\ S\theta_4 & C\theta_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^4T_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

最后得出，末端执行器相对于基座的变换矩阵

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 = \begin{pmatrix} \cos(\theta_1 + \theta_2 + \theta_4) & -\sin(\theta_1 + \theta_2 + \theta_4) & 0 & a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2 + \theta_4) & \cos(\theta_1 + \theta_2 + \theta_4) & 0 & a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & d_3 + d_5 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Z



设出相应参数，建立坐标系，列出参数表

	a_{i-1}	a_i	d_i	θ_i
1	0	0	0	θ_1
2	90°	a_1	d_2	0
3	90°	0	0	θ_3
4	0	a_3	0	0

得出变换矩阵

$${}^0T_1 = \begin{pmatrix} C\theta_1 & -S\theta_1 & 0 & 0 \\ S\theta_1 & C\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^1T_2 = \begin{pmatrix} 1 & 0 & 0 & a_1 \\ 0 & 0 & -1 & -d_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2T_3 = \begin{pmatrix} C\theta_3 & -S\theta_3 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ S\theta_3 & C\theta_3 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad {}^3T_4 = \begin{pmatrix} 1 & 0 & 0 & a_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0T_4 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 = \begin{pmatrix} \cos(\theta_1 - \theta_3) & \sin(\theta_1 - \theta_3) & 0 & a_1 \cos \theta_1 + a_3 \cos(\theta_1 - \theta_3) + d_2 \sin \theta_1 \\ \sin(\theta_1 - \theta_3) & -\cos(\theta_1 - \theta_3) & 0 & a_1 \sin \theta_1 + a_3 \sin(\theta_1 - \theta_3) - d_2 \cos \theta_1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

注意：要为末端单独建立参考系，该参考系的选取可以和最后一个关节重合，但是必须要体现出来